EXPERIMENT: ANALYSING THE BEHAVIOUR OF NETWORK DEVICES USING CISCO PACKET TRACER SIMULATOR

AIM:

To analyse the behaviour of PCs, a switch and a router using Cisco Packet Tracer by configuring IP addresses, testing connectivity and observing ARP resolution, MAC-address learning and router interface status.

REQUIREMENTS:

Cisco Packet Tracer, two PCs, one Layer 2 switch, one router and copper straight-through Ethernet cables.

PROCEDURE:

1. Open Cisco Packet Tracer and create a new project.

2. Place two PCs, one switch and one router in the workspace.

3. Connect PC0 and PC1 to separate switch ports. Connect the switch to the router's GigabitEthernet0/0 interface using suitable Ethernet connections.

4. Configure PC0 with IP address 192.168.1.10, subnet mask 255.255.255.0 and default gateway 192.168.1.1.

5. Configure PC1 with IP address 192.168.1.11, subnet mask 255.255.255.0 and default gateway 192.168.1.1.

6. Configure the router interface with IP address 192.168.1.1 and subnet mask 255.255.255.0. Enter the no shutdown command.

7. From the PC command prompt, ping the router and the other PC.

8. Use the switch and router CLI show commands to inspect the MAC address table and interface status.

9. Select Simulation mode, filter for ARP and ICMP, and use Capture/Forward to observe packet movement.

10. Record the output and save the Packet Tracer project.

IP ADDRESS CONFIGURATION:

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	192.168.1.1
Router G0/0	192.168.1.1	255.255.255.0	—

OUTPUT:

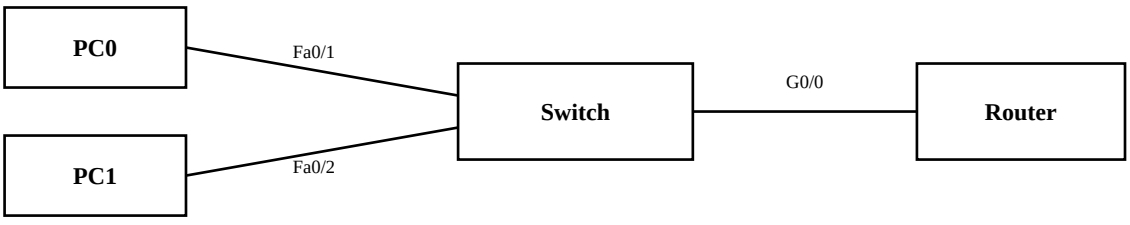

Fig. 1. Simulated network topology of two PCs, a switch and a router.

Router CLI configuration:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end
```

```
Router# show ip interface brief
```

Sample router output:

```
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
```

OUTPUT:

PC0 command prompt – sample successful connectivity test:

```
C:\> ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

PC0 to PC1 connectivity test:

```
C:\> ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Switch CLI – sample MAC address table:

```
Switch> enable
Switch# show mac address-table

          Mac Address Table
-------------------------------------------
Vlan    Mac Address       Type       Ports
----    -----------       --------   -----
  1     00D0.BA12.1001    DYNAMIC    Fa0/1
  1     00E0.F7A4.2202    DYNAMIC    Fa0/2
  1     00A1.B2C3.D4E5    DYNAMIC    Gi0/1
```

Note: MAC addresses above are illustrative sample values. Actual addresses and port names depend on the Packet Tracer devices and connections.

OBSERVATION:

1. The router interface becomes up/up when the interface is enabled and the physical/virtual link is connected.

2. Successful ping replies indicate IP connectivity between the source and destination devices.

3. ARP resolves an IPv4 address to a MAC address before a device sends a frame to a local destination.

4. The switch learns source MAC addresses dynamically and records the associated switch ports.

5. In Simulation mode, ARP request/reply events may appear before the first ICMP echo request/reply.

RESULT:

Thus, the behaviour of network devices was analysed using Cisco Packet Tracer. IP addresses were configured, connectivity was verified using ping, and ARP resolution, switch MAC-address learning and router interface status were studied.

Note: The command outputs are sample expected outputs for the configured topology. Replace or supplement them with screenshots captured from your own Packet Tracer session if your lab record requires actual execution evidence.